# *Africanis*: the world's last pure breed of dog? 29A

The *Africanis*, a breed of dog native to Africa for over 70 centuries, is believed to be the last remaining pure breed of dog, in the truest meaning of the term 'pure bred'. It possesses features of a breed whose gene pool has been directed by natural selection alone and not by human intervention. As for all pure-bred animals, individuals share similarities. In the case of the *Africanis* these are:

- a related gene pool
- overall head and body shape
- short fur
- medium size
- friendly, intelligent and alert personality
- fitness, active behaviour and general good health.

**Dogs:** carnivorous mammals of the family *Canidae*. There are about 40 species, which can range from 1.5 kg to 75 kg when fully grown.

Because natural selection has acted on the *Africanis* gene pool, it remains richly varied. For this reason, although the overall appearance of the breed is recognisable, each dog has individual characteristics. These differences include different:

- colouring
- markings (solid colour or multiple colour markings).

These differences make the breed particularly attractive to humans because a choice of 'models' is available. People can select a look that most appeals to them. In order to safeguard the future of the *Africanis*, it has recently been officially recognised as a breed in the hopes it will not be interfered with or left to interbreed with other types of dog, producing mongrels.

**A mongrel:** an individual resulting from the interbreeding of different breeds.

Write brief definitions for the following terms with relation to their meaning in genetics.

1. enhancing
2. side-effects
3. native
4. heterogenous
5. mongrel
6. species
7. breed

8. Many people in Africa consider that the *Africanis* is fitter and has a more balanced nature than many other breeds of dog. Give one reason why this breed might be 'superior' to other breeds.

9. Using a variety of sources (text books, conventional encyclopaedias, CD ROMs, the Internet, etc.) research the *Africanis* and prepare a report on it. Your teacher will tell you whether to use an oral presentation, a poster, an assignment, etc. If using the Internet, try: Kennel Union of South Africa or Africanis Society of Southern Africa.

## Genetics and dog breeding

The term 'pure-bred' refers to an animal that is homozygous for a gene or set of genes. For example, a pair of pure-bred Irish Setters will produce a litter of similar puppies closely resembling their parents. Pure-bred dogs are bred for specific features, such as size, colour, personality, appearance, etc. Different breeds can become fashionable, simply based on these characteristics. Unwanted side-effects can sometimes be linked to features wanted by breeders. These can occur due to the enhancing of features beyond normal, healthy limits or as a result of desired features being genetically linked to a negative (unwanted) trait.



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29B

Think about the various breeds of dogs that exist. People own dogs for various reasons, such as companionship, or to help with work, etc.

**10** Choose three examples of dog breeds that stand out in your mind. State the image/stereotype associated with each dog. Describe the type of person that might choose each of the dogs described.

Enhancing (increasing) certain features can sometimes lead to unwanted complications. Many 'toy' breeds produce litters with very small puppies, having high mortality rates among the puppies.

**11** Describe what is meant by a 'toy breed'. Give an example.

**12** What is meant by the term 'high mortality rate'?

**13** Why might some toy breed females have difficulty in producing live young?

**14** Give three factors which contribute to the cost of pure-bred puppies.

**15** Describe the possible positive (good) and negative (bad) effects resulting from selective breeding for the following traits:

- short, flat snout (nose)
- long body
- narrow snout
- chunky (compact) body, short legs
- loose skin, many folds
- protective feelings for owner/property.

Selective breeding can produce unwanted features due to 'genetic linkage'. For example a certain dog breed was once popular because it was said to be safe around young children. A new colour was produced by a rarely occurring recessive mutation (unexpected genetic change). This attractive colour quickly became popular and many litters were produced. A 'brindle' variation followed, by breeding animals of the new colour with animals of standard colours.

**Brindled:** dark streaks or flecks on a grey or tawny background.

Over some years, and after a few regrettable incidents, people observed a link between the recessive mutation, the new colouration and aggression. This new, unexpected and unpredictable dangerous behaviour made the breed less attractive to couples with young children. A once reliable breed had become potentially dangerous, and many people are now reluctant to leave dogs of this type alone with young children.

**16** Study the following paragraph and explain the words printed in **bold** type.

A **commercial** breeder of this type of dog said, 'The breed has been **contaminated**. Some **brindle** examples of our dogs could now be considered **unreliable.** The well known original coloured **pure-bred** dog is very safe. The **mutation** has given rise to **aggressive** behaviour. But even some dogs that don't exhibit any trace of the new colour can be considered unreliable because the characteristic is **recessive**.'

**17** Prepare a report/presentation about a pure-breed of dog of your choice. Include:

- special features for which the breed is known, including a photo/illustration of an example of the breed
- the breed's country of origin and how long the breed has been in existence
- the reputation of the breed (for example what sort of people choose it, the type of image it projects)
- the advantages and disadvantages that may have resulted from selective breeding in the breed.

If possible, contact a breeder for information about the breed. Compare this information during a class discussion.

*Links have been made between defective pigmentation genes and vision. The gene that produces the Siamese cat's unique colouration is linked to crossed eyes!*





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